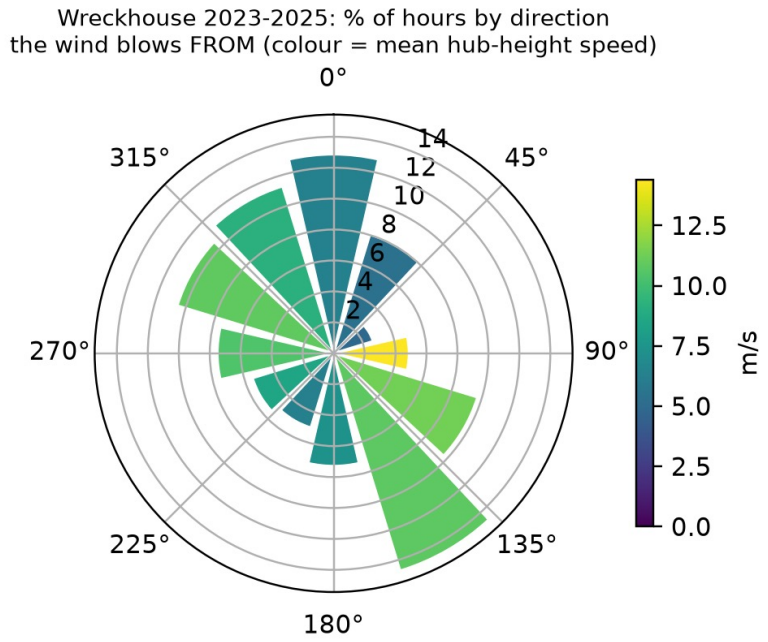
# WakeWise NL

Where should wind turbines go on Newfoundland's west coast so they don't steal each other's wind? Real wind, real terrain, QAOA in Qiskit, and up to 24 qubits on IBM's quantum computer in Quebec

**Team AI Entanglers**

Mostafa Wasib

Qiskit Fall Fest 2026 · Memorial University




Design wind: 26,264 hours at Wreckhouse, NL, 2023-2025; we test on all 19 years, 2007-2025. Longer petals: directions the wind blows from more often.

# WakeWise NL in one minute

## 154 GWh/yr

from the 9-turbine layout we recommend on our flat test parcel: the electricity of ~9,100 NL homes

## ~345x

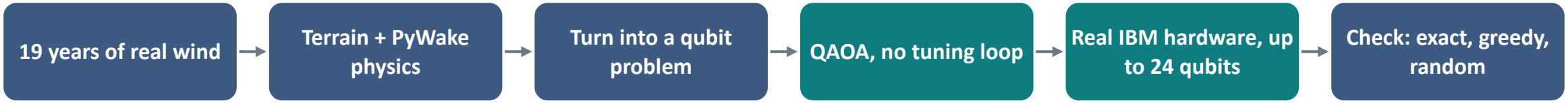
more often than random guessing, QAOA returns the best layout (flat parcel, 12 qubits, simulation)

## 24 qubits

real IBM hardware, real terrain: QAOA gets 66% of the way from random guessing to the best layout; the exact best appeared (chance: 1 in 17 million)

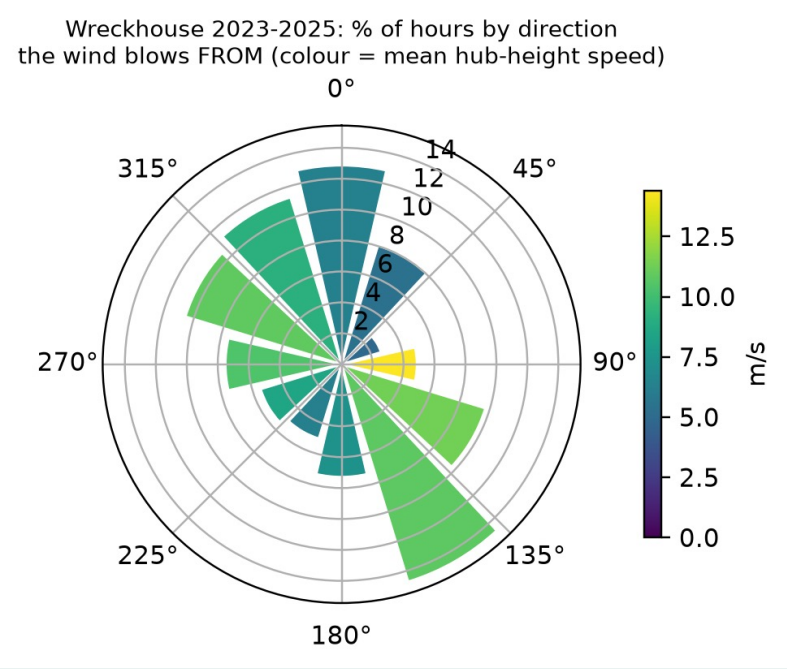
## 14 of 19 years

real wind years (2007-2025) in which the flat parcel's recommended layout is the best possible; a one-year design: 5 of 19




*Teal = quantum steps. No quantum advantage is claimed: at this size we can check every answer exactly.*

# Strong NL wind, and the wakes that waste it




Design wind: 26,264 hours at Wreckhouse, 2023-2025; we test on all 19 years, 2007-2025. The wind comes mostly from the south-east and north-west, so turbines lined up on that axis shade each other most.

Sources: ECCC hourly climate data (Wreckhouse, ID 8404343); NL Government EA bulletin, 9 Apr 2024; CBC News; CREATE-SWIFT talk, Qiskit Fall Fest 2026.

### Wreckhouse, southwest Newfoundland

Gales here once blew trains off the tracks. At turbine height (110 m) the wind averaged 9.0 m/s over 2023-2025, so a 4 MW turbine would average 50% of its full output.

### Project Nujio'qonik, next door

150+ turbines proposed in the Codroy Valley and on the Port au Port Peninsula to make hydrogen in Stephenville. Released from provincial environmental assessment, with conditions, in April 2024.

### MUN's CREATE-SWIFT program asks for this

Compare classical and quantum methods on wind-energy optimization: accuracy, cost, scalability and real-world relevance.

**17,600 MWh/yr**

from one 4 MW turbine standing alone (2023-2025 wind, DTU PyWake)

**up to 2.4%**

lost to the wake (slower air) of one badly placed neighbour; a block of 12 loses 4.2%

**32% of hours**

at full power anyway, when a wake costs almost nothing: wakes matter less here than at most sites

**Sustainability benefit:** more clean electricity or green hydrogen from the same land and steel. The prototype shows the method; real benefit needs site-specific terrain, turbines and costs.

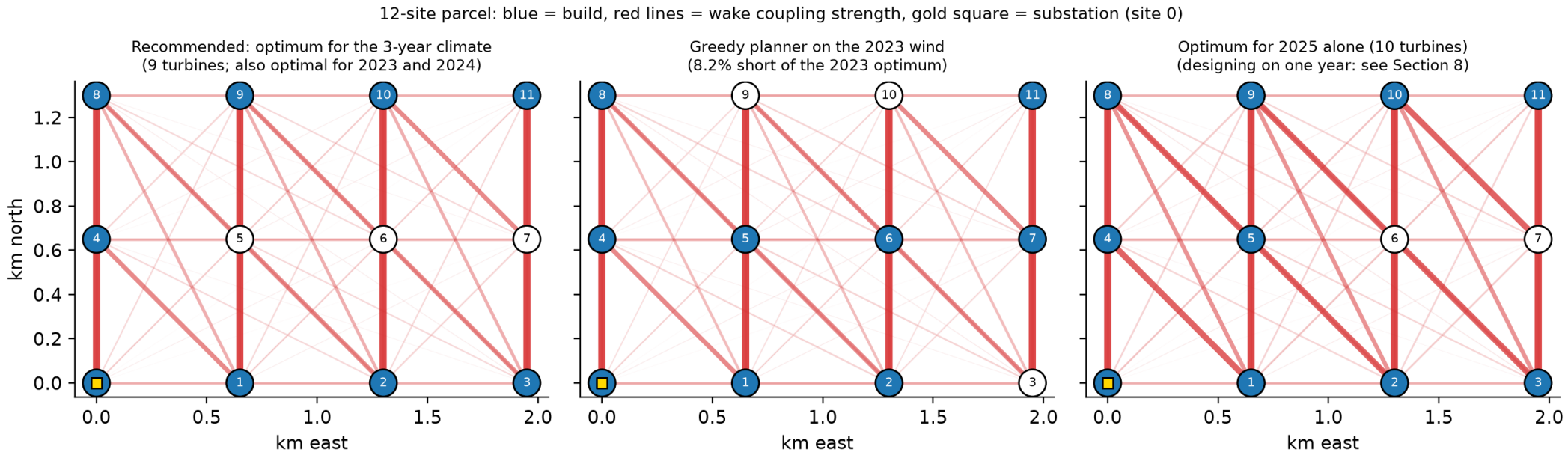
# Inputs, output and scope

| Input | What we used | Status |
|---|---|---|
| Wind | Wreckhouse weather station, hourly: 2023-2025 to design (26,264 h), 2007-2025 to test (163,068 h); scaled from 10 m up to the 110 m turbine height | **Measured (height scaling assumed)** |
| Wake physics | DTU PyWake, an industry-standard simulator; cross-checked with full-farm runs and a second wake model | **Industry-standard model** |
| Turbine | Generic 4 MW turbine with a 130 m rotor (PyWake's reference curves) | Illustrative |
| Costs | Each turbine's yearly cost, counted in energy: 88% of what a lone turbine makes, plus 2% per 650 m of cable; other levels tested | Illustrative, tested |
| Real-terrain parcel | Satellite elevation (Copernicus) and DTU's Global Wind Atlas; sites picked by a fixed rule | **Measured + modelled (screening)** |
| Output and use | One layout: build (1) or don't build (0) at each site. We report the best layouts, the energy at stake and whether the choice holds up year to year; for early screening, not construction advice | |
| Scope | Flat test parcel: 12-16 sites on a 650 m grid, 19 years, a wetland case. Real-terrain parcel: 12-24 sites. Simulation, noise model and real hardware | |

# Wakes make this a puzzle, not a checklist

$$f(x) = -\Sigma\,(E - c_i)\,x_i \;+\; \Sigma\,W_{ij}\,x_i\,x_j$$

Each turbine earns its energy E minus its cost $c_i$; each built pair loses $W_{ij}$ to wakes. A QUBO: yes/no choices with pair terms; lower is better




**A simple greedy rule (keep adding the best next turbine) finds the 3-year optimum, but misses the best layout in 16 of 19 single years (up to 8.2% short). We check every answer against all 4,096 possible layouts.**

# Wake physics maps straight onto qubits

$$x_i = (1 - z_i) / 2$$

$$H = \Sigma\, h_i\, Z_i \;+\; \Sigma\, J_{ij}\, Z_i\, Z_j$$

Each site is a qubit. A turbine's value acts on its own qubit (a Z field); a wake loss links two qubits (a ZZ coupling). This is an Ising model.

## Why QAOA fits

- Already yes/no with pair terms: no extra qubits, no penalty tricks
- Grows fast: 24 sites give 16.8 million layouts (NP-hard in general)
- Tuned on the best 25% of samples (CVaR): a planner needs the best layout, not the average
- Small enough to check exactly, so we measure quality instead of assuming it

1. **Map** *classical*
   Wind and PyWake → QUBO → Qiskit QAOA circuit (qaoa_ansatz), 1 to 3 layers
2. **Optimize** *classical*
   Compile for IBM's chip; drop the weakest wake links to save two-qubit gates
3. **Execute** *quantum*
   Angles tuned in simulation, or predicted or reused at 20-24 qubits; run on ibm_quebec
4. **Post-process** *classical*
   Score every sampled layout; compare with exact, greedy and random

# QAOA works in simulation and on real hardware

**Noiseless simulation, 12 qubits**

## 8.4% vs 1 in 4,096

of shots give the best layout: QAOA with 3 layers vs random guessing, about 345 times more often

## 34 vs 12,269

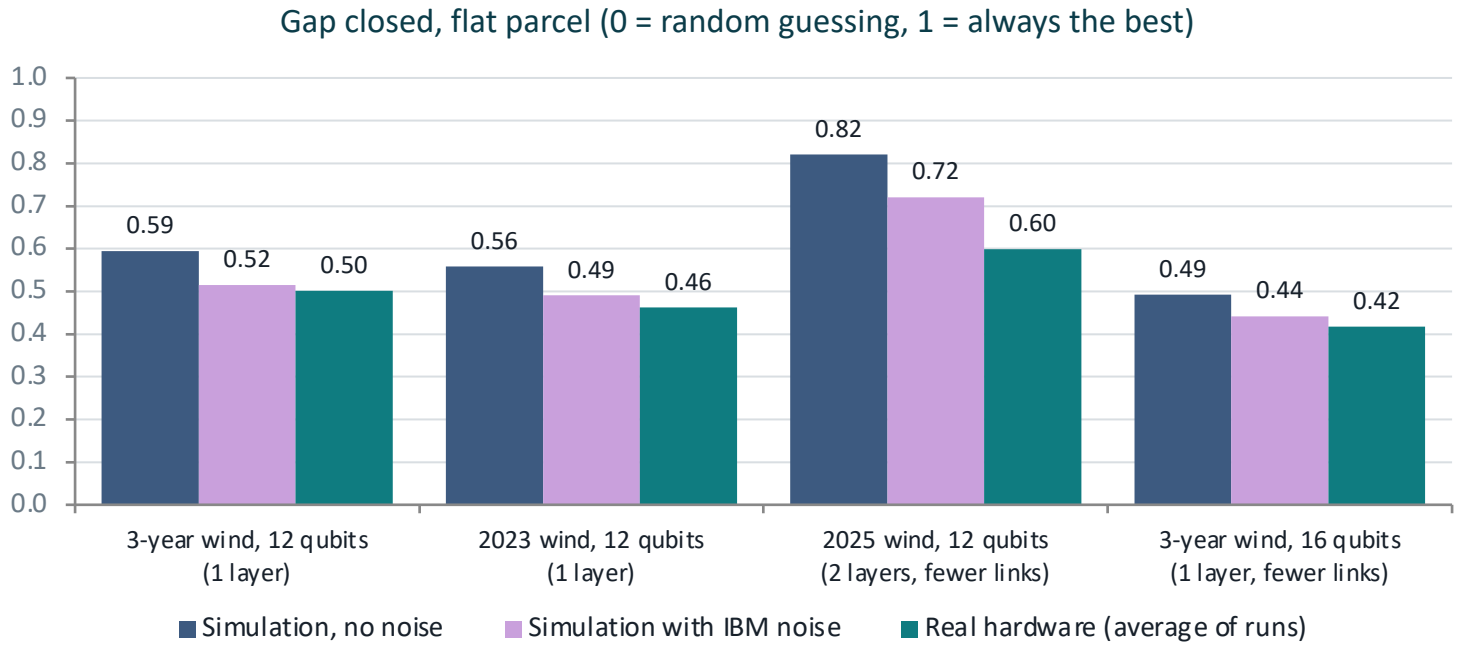
shots needed to be 95% sure of seeing it (1 layer: 388; 2 layers: 54). One shot = one sampled layout

## 20%

of QAOA samples (2 layers) beat the greedy rule in the 16 years where greedy misses

*Honest caveat: a typical QAOA sample is good but not the best; the value is in its best few dozen shots.*

**Real hardware: ibm_quebec**




ibm_quebec: IBM Quantum System One in Bromont, Quebec, 7 Nov 2026.

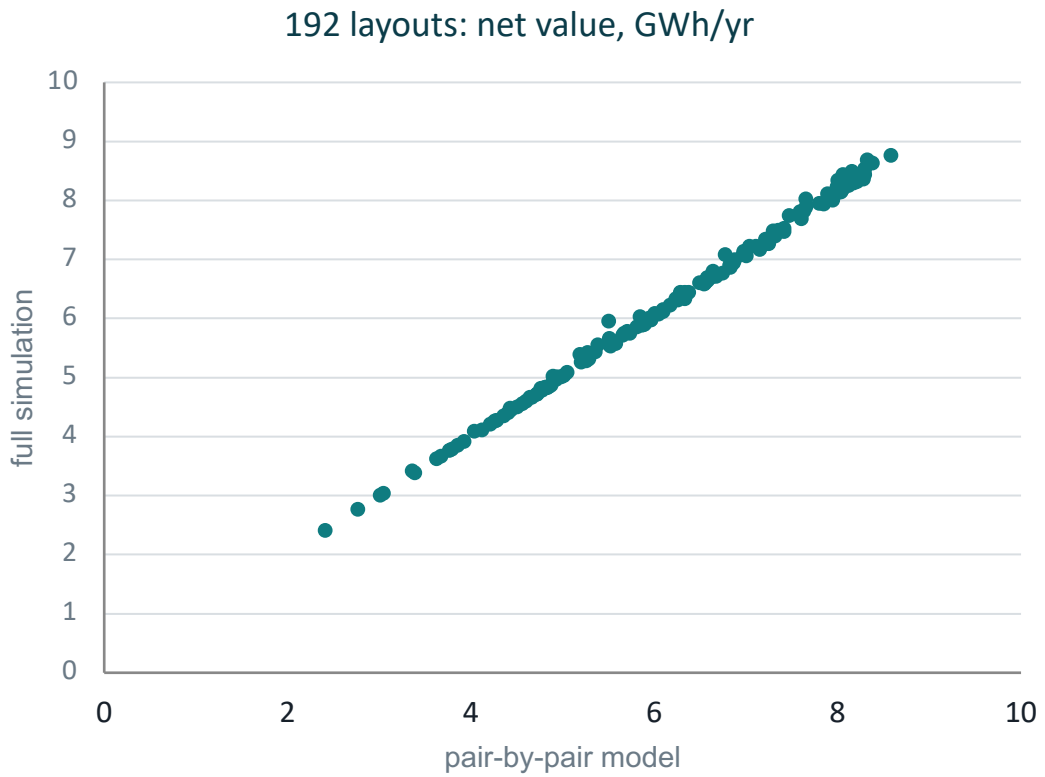
Main case, 4 runs: the best layout 23 times more often than random guessing.

**2023 wind: 15% of hardware samples beat the greedy rule (random guessing: 2%).**

A control circuit with no problem in it scores -0.02, the same as random guessing: the gain is QAOA, not noise.

We found a faulty qubit (101): it misread 1 as 0 in 65% of shots, though IBM's data looked normal. We now keep such qubits out of every run.

# Validated: full simulations and 19 years of wind




Our pair-by-pair model ranks layouts in nearly the same order as full simulations (rank correlation 0.993; 1 = same order) and as a second wake model (0.992). Its best layout is the best in both.

| Layout designed on | Best in | Avg. shortfall | Worst-year shortfall |
|---|---|---|---|
| **2023-2025 (recommended)** | **14 of 19** | **19 MWh/yr** | **167 MWh/yr** |
| 2025 wind only | 5 of 19 | 104 MWh/yr | 416 MWh/yr |

## 14 of 19 years

the recommended layout is the best possible; one designed on 2025 alone: 5 of 19

Wind changes from year to year, so we test each design on every year back to 2007. Shortfall = energy lost in a year compared with that year's best layout.

# Real-world limits: a wetland and real terrain

## Protecting a wetland

*Illustrative: two sites inside a buffer around the Grand Codroy Estuary, Newfoundland's only wetland of international importance (a Ramsar site).*

### 519 MWh/yr

of net energy: the price of keeping turbines out of the wetland

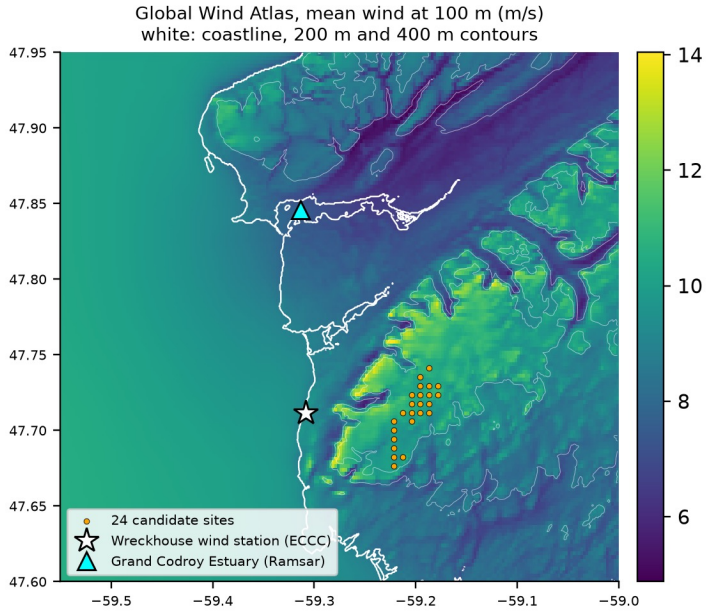
### 9 → 8 turbines

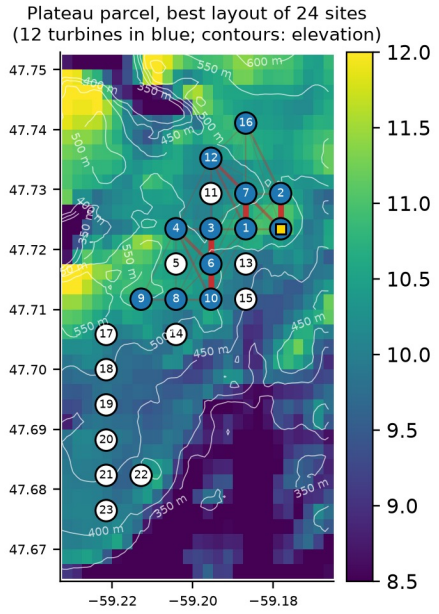
the optimizer re-plans the whole layout, not just drops two sites

### 133x random

QAOA (3 layers, simulation) finds the new best layout in 3.2% of shots: just a penalty on two qubits

**Real terrain: screening level**




### 423-550 m

above sea level: 24 buildable sites on the plateau above Wreckhouse, with 9.7-10.9 m/s average wind in the atlas

### 12 of 19 years

best possible when designed on all 19 years; a 2023-2025 design: 4 of 19. On terrain, use the long record

### 804x

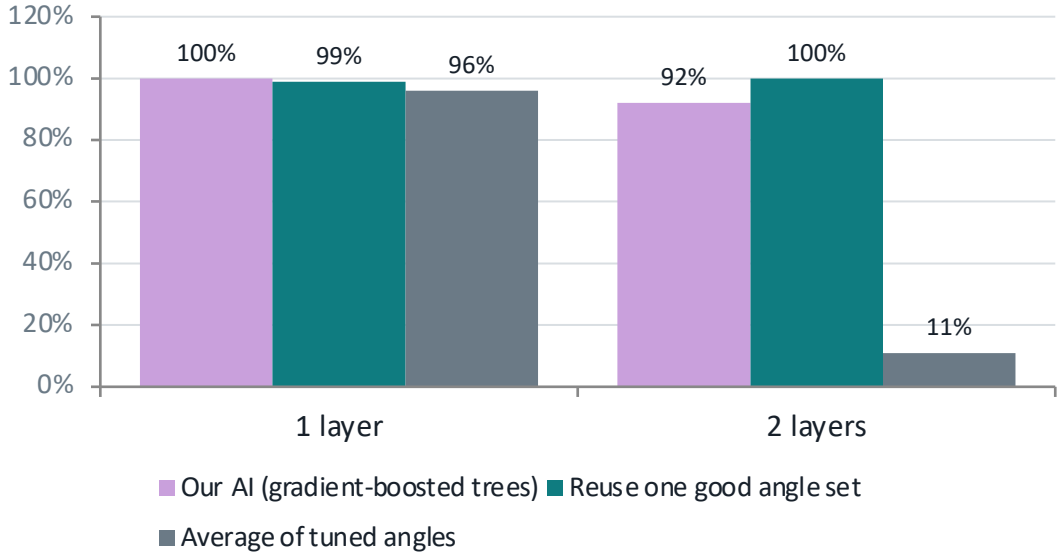
more often than random guessing: QAOA (2 layers, simulation) finds the best layout for the first 16 sites

*Site rule: on land, slopes up to 10%, terrain like the weather station's (so the wind atlas can be trusted), at least 3 km from the Ramsar estuary, growing out from the windiest spot. Wind: the station's real hours, scaled per site by DTU's Global Wind Atlas.*

# Scaling to 24 qubits with no tuning loop

**Simulation, unseen years 2020-2025**

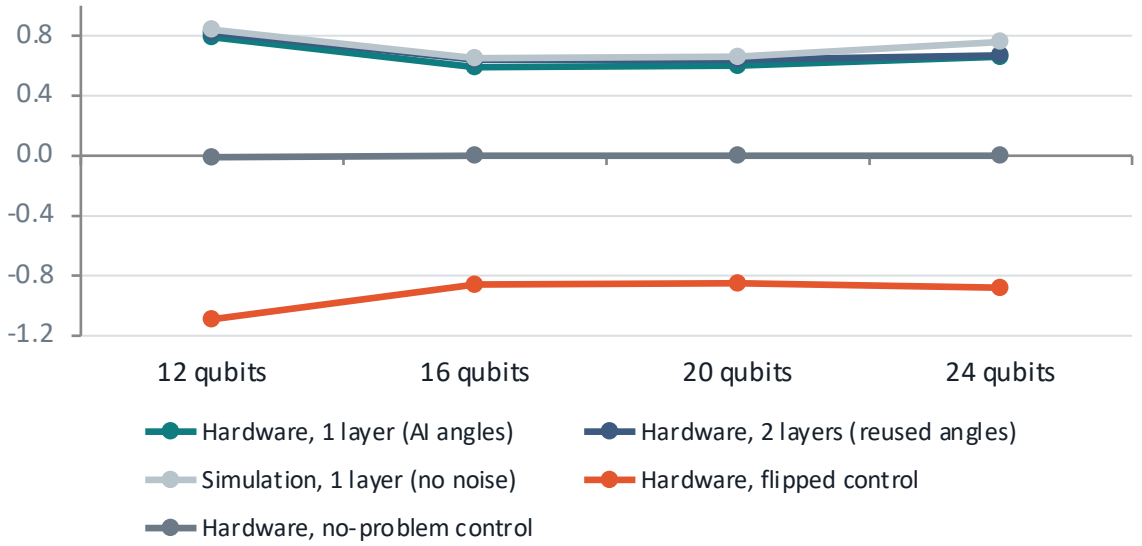
Share of fully tuned quality (90 test cases)




- **100% / 92%**: our AI with 1 / 2 layers, no tuning loop needed
- **99% / 100%**: reusing one good angle set beats our AI, an honest negative result
- **11%**: averaging tuned angles fails at 2 layers; good angles differ, so their average lands in between

**Real hardware: ibm_quebec**

Gap closed on real terrain (0 = random, 1 = always the best)




- **0.66 vs 0.76**: gap closed at 24 qubits, real hardware vs noiseless simulation
- **4 in 24,000** shots gave the exact best of 17 million layouts (random guessing: 1 in 17 million)
- **-0.88**: flipped control, the same gates with one sign reversed, lands below random. The machine follows the physics

# What the evidence supports, and what it does not

## Supports

- Siting on real NL wind is a real puzzle; our pair-by-pair model matches full simulations
- The recommended layout is the best possible in 14 of 19 real wind years
- QAOA favours the best layouts: ~345x random in simulation, and a clear signal on real hardware at 24 qubits
- No tuning loop needed at 20-24 qubits: our AI works, and reusing one angle set works as well
- Control circuits behave as physics predicts: the flipped circuit scores below random

## Does not support

- Quantum advantage: a laptop still solves 24 sites exactly
- Bankable siting: the terrain parcel is a screening study, and costs are illustrative
- That our AI beats simply reusing one good angle set: it does not
- That the faulty qubit caused the weaker 2-layer results on the flat parcel: on healthy qubits they were no better

**Next:** measure wind on the plateau · add spacing and habitat limits · test bigger parcels against strong classical solvers (simulated annealing, MILP) · partner through MUN CREATE-SWIFT and Mitacs

**Builds on** Hancock et al. (2023) and Nigro et al. (2025), who set out quantum wind-farm layout and compare solvers. New here: a real NL site with 19 years of measured wind and real terrain, a check against full wind-farm simulations, and controlled hardware runs up to 24 qubits.

*Resources: notebook about 30 minutes on a laptop; 21 hardware jobs, 174 s of quantum-computer time in total.*

**Thank you. Questions?**
**github.com/mrw-soumik/Qiskit-AI-Entanglers-**

Credits: QAOA workflow adapted from Tinh Bui's workshop notebook; IBM Qiskit 101 and the Fall Fest hardware example; DTU PyWake; ECCC data (Open Government Licence - Canada); DTU Global Wind Atlas (CC BY 4.0); Copernicus DEM (ESA); prior quantum wind-farm layout studies: Hancock et al. (2023), Nigro et al. (2025). Full credits in the notebook.